

Technical Portfolio - Scenario 4

Group 3

Circuit Components

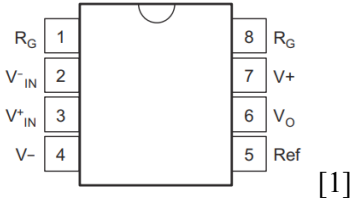
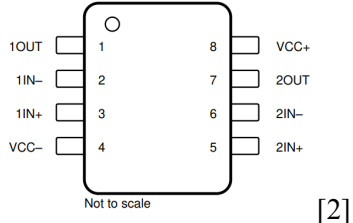
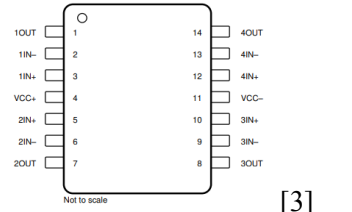
Blocks	Instrumentation Amplifier	Active Rectifier	LPF, HPF, and Non-inverting amplifier
Pinout			
Op-amp	INA128	TL082	TL084

Table showing the op-amps used in the circuit

The circuit required the use of six op-amps but only 3 physical op-amps were used for efficiency and cost reduction reasons. For the instrumentation amplifier, INA 128 was employed. The active rectifier needed two operational amplifiers, so TL082 was chosen because it has two channels. TL084 was used instead of three different op-amps for the sallen-key low pass, sallen-key high pass filters and the non-inverting amplifier.

General Circuit Diagram

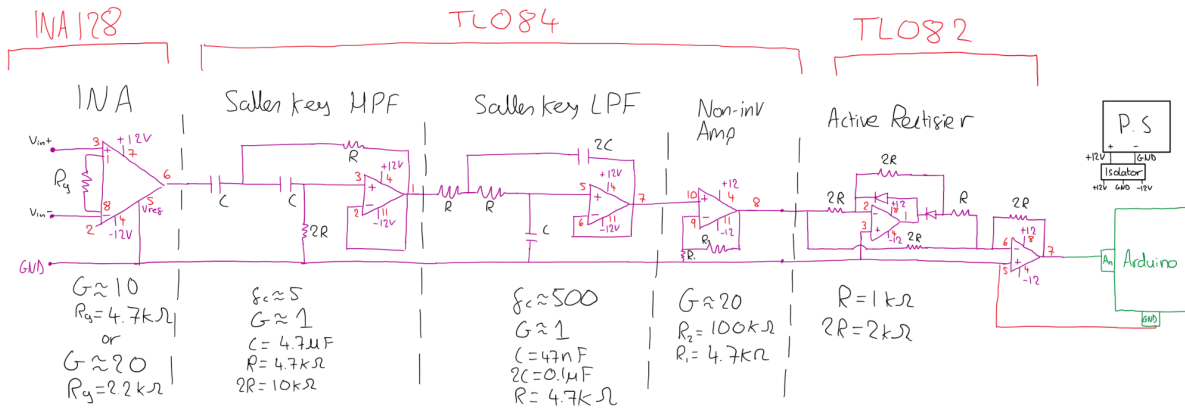


Figure 1. Circuit Diagram of the EMG processing circuit

- Instrumental amplifier - Takes the difference of 2 input terminals and amplifies
- Sallen-key High pass filter - Attenuates low frequencies below a cutoff frequency (5Hz) [4]
- Sallen-key Low pass filter - Attenuates high frequencies above a cutoff frequency (500Hz) [4]
- Non-inverting amplifier - Amplifies signal after filtering (Gain = 10 or 20 depending on muscle)
- Active full-wave rectifier - Outputs entire signal in 0-5V range with reduced signal loss

Filter plots

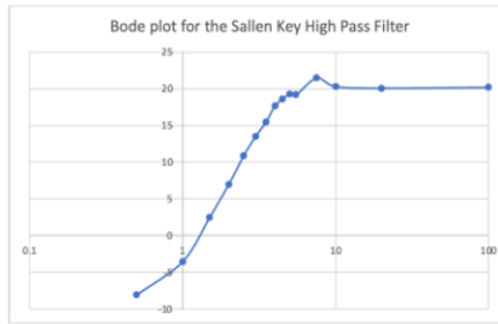


Figure 2. Characterisation of the Sallen-Key HPF

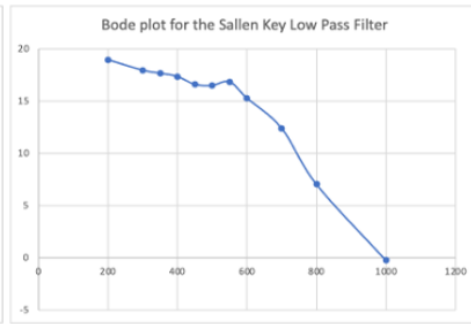


Figure 3. Characterisation of the Sallen-Key LPF

Circuit layout for each degree of freedom

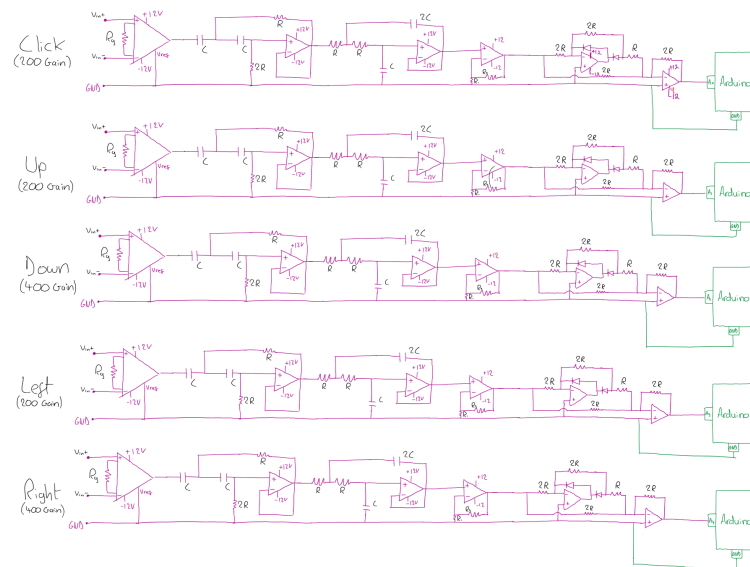


Figure 4. Circuit diagram of the five degrees of freedom

Final Circuit

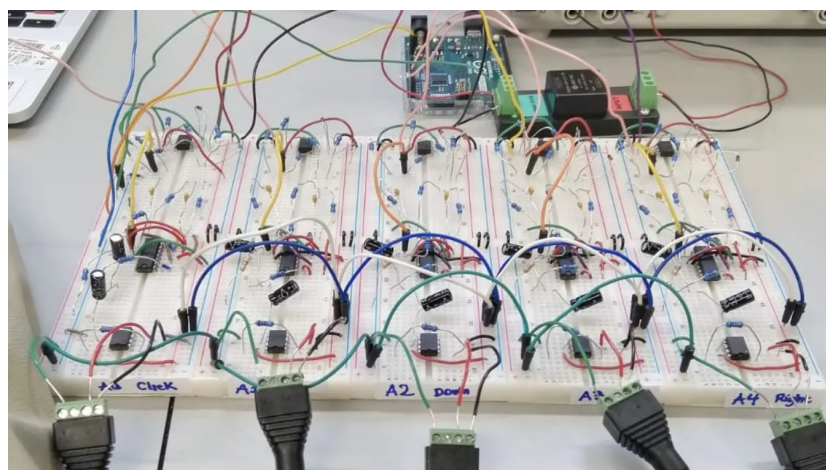


Figure 5. Picture of the circuit (5 degrees of freedom)

Configuring thresholds from signals

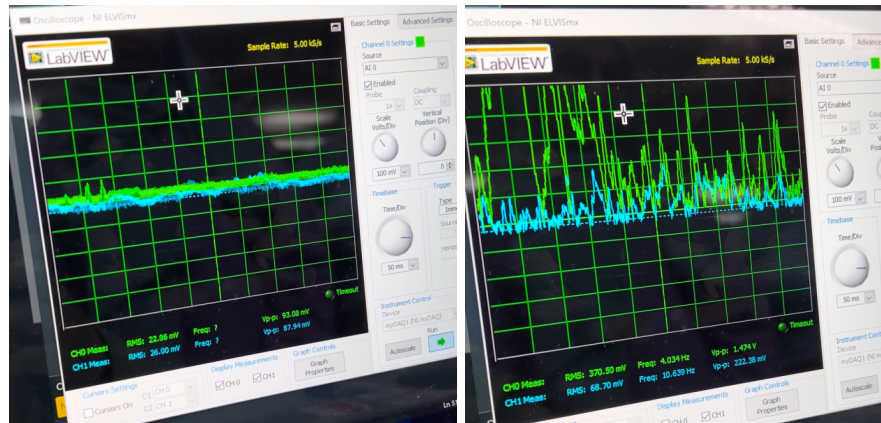


Figure 6. Determination of thresholds using myDaq oscilloscope (example of UP movement)

The image on the left shows muscles at rest, and on the right is the “up” movement. The blue signal is the triceps brachii and the green one is the biceps brachii. The green signal is much higher than the blue one, which allows us to determine different thresholds for different speeds in the code.

Electrode position

- Location of electrodes (Figure 7). This was chosen based on the 5 physiological movements: flexion for up, extension for down, clenching for click, supination of forearm for right, and shoulder extension for left.
- Distance between electrodes: initially started with a distance of 18 to 36 mm (around 2 to 4 cm) [5], then proceeded with trial and error until high enough signals were achieved.
- Direction of electrodes: longitudinal following the orientation of the muscle fibre (this is the direction in which the action potential travels) [6].

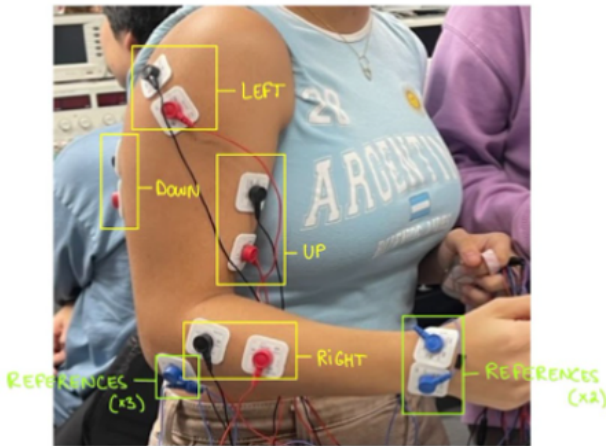


Figure 7. Image showing the electrodes position

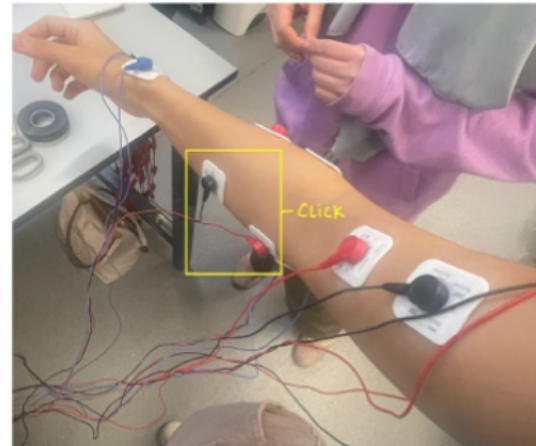


Figure 8. Image showing the electrodes position (inner arm)

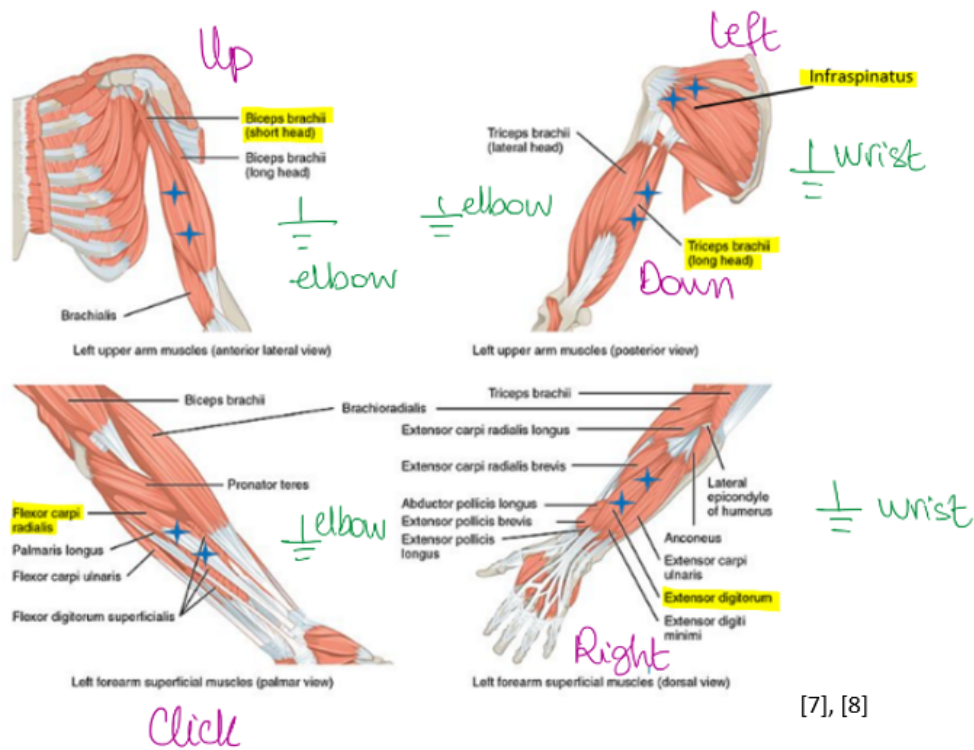
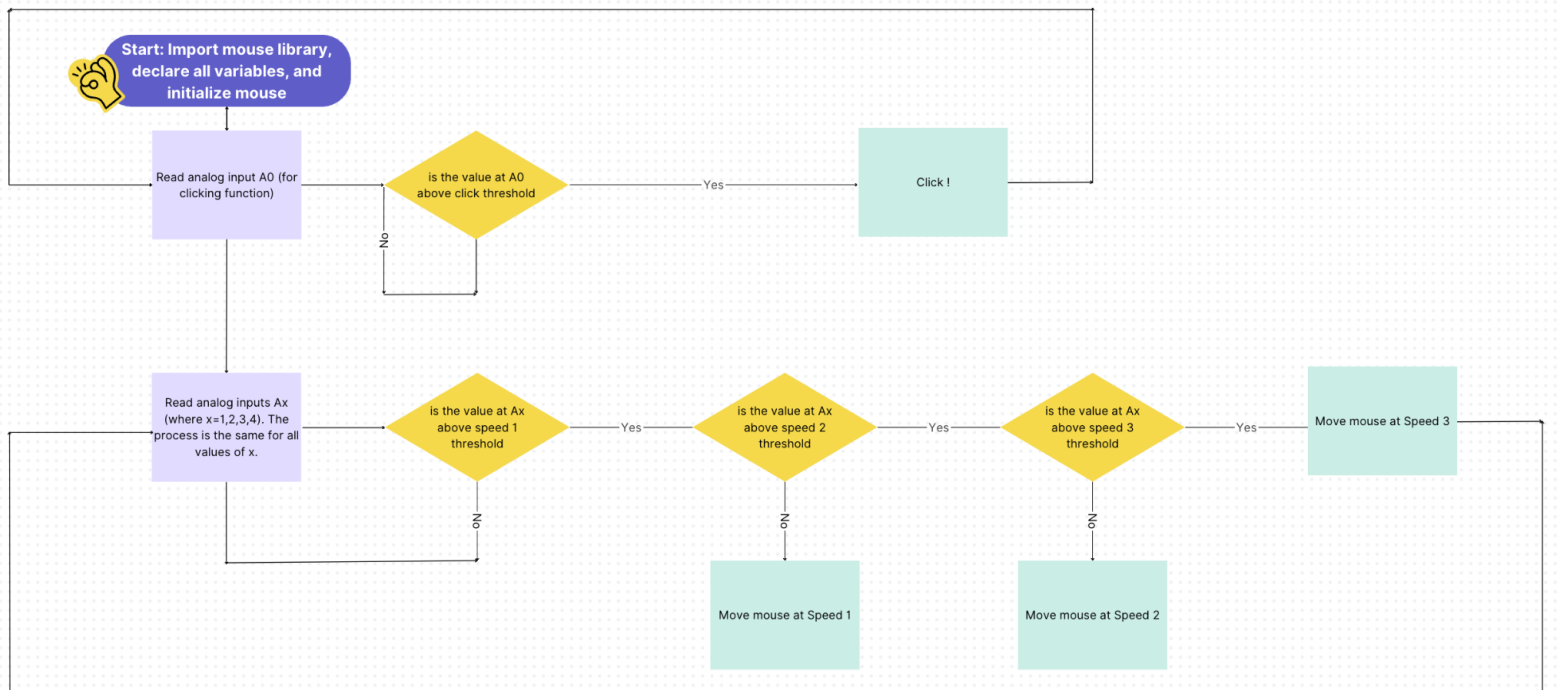


Figure 9. Diagram of the muscles and electrodes position

Coding flowchart



Appendix

Code

The final code used to describe the control of the 5 degrees of freedom. It is a combination of all the sub-codes used for the individual movements. [9]

```
#include <Mouse.h>           //import library which has mouse functions

int analogPin = A0; //assign A0 for Click input
int val0 = 0;        // variable which stores readings of A0
int analogPin1 = A1; //assign A1 for moving UP input
int val1 = 0;        // variable which stores readings of A1
int analogPin2 = A2; //assign A2 for moving DOWN input
int val2 = 0;        // variable which stores reading of A2
int analogPin3 = A3; //assign A3 for moving LEFT input
int val3 = 0;        // variable which stores reading of A3
int analogPin4 = A4; //assign A4 for moving RIGHT input
int val4 = 0;        // variable which stores reading of A4
char xAxis = 0;      //x axis set to 0, no movement will be done in that direction when
this variable is declared
char wheel = 0;      //scroll wheel set to zero because it was not needed in the scope
of this project
char yAxis1 = -15; //up speed1 (number of pixels it increments by, negative sign to
make it go upwards)
char yAxis2 = -80; //up speed2
char yAxis3 = -100; //up speed3
char yAxis4 = 15; //down speed1
char yAxis5 = 80; //down speed2
char yAxis6 = 100; //down speed3
char yAxis = 0; //y axis set to 0, no movement will be done in that direction when
this variable is declared
char xAxis1 = -15; //left speed1
char xAxis2 = -80; //left speed2
char xAxis3 = -100; //left speed3
char xAxis4 = 15; //right speed1
char xAxis5 = 80; //right speed2
char xAxis6 = 100; //right speed3
void setup() {
  Serial.begin(9600);
  Mouse.begin();           //initialize the mouse
}
```



```
void loop() {  
  //CLICK//  
  val0 = analogRead(A0);      // read the signal on A0  
  if (val0>13) {              //threshold was found using myDaq oscilloscope signal values  
  
    Mouse.click(); //left Click mouse once a threshold in A0 is met  
    delay(500); //delay allows click to just happen once  
  }  
  
  //UP AND DOWN CODE//  
  val1 = analogRead(A1);  
  //MOVING UP//  
  if (val1>110) {  
  
    Mouse.move(xAxis, yAxis1,wheel); //move up at speed1 once threshold met  
    delay(50); //moves only once, slightly longer delay so that mouse moves slower  
  
  } else if(val1>160) {  
  
    Mouse.move(xAxis, yAxis2,wheel); //move up at speed2 once threshold met  
    delay(75); //moves only once  
  
  } else if(val1>170) {  
  
    Mouse.move(xAxis, yAxis3,wheel); //move up at speed3 once threshold met  
    delay(10); //as speed increases, we want a smaller delay to allow the mouse to move  
faster and in a fluid way  
  
  }  
  //MOVING DOWN//  
  val2 = analogRead(A2);      //similar pattern and explanations as moving up for moving  
down, left and right  
  
  if (val2>140) {  
  
    Mouse.move(xAxis, yAxis4,wheel); //Move down at speed1 once threshold met  
    delay(50);  
  
  } else if(val2>160) {  
  
    Mouse.move(xAxis, yAxis5,wheel); //Move down at speed2 once threshold met
```

```
delay(75);

} else if(val2>170) {

Mouse.move(xAxis, yAxis6,wheel); //Move down at speed3 once threshold met
delay(10);
}

//LEFT AND RIGHT CODE//
val3 = analogRead(A3);

//MOVING LEFT//
if (val3>80) {

Mouse.move(xAxis1, yAxis,wheel); //Move left at speed1 once threshold met
delay(50);

} else if(val3>95) {

Mouse.move(xAxis2, yAxis,wheel); //Move left at speed2 once threshold met
delay(75);

} else if(val3>120) {

Mouse.move(xAxis3, yAxis,wheel); //Move left at speed3 once threshold met
delay(10);

}
//MOVING RIGHT//
val4 = analogRead(A4);
if (val4>60) {

Mouse.move(xAxis4, yAxis,wheel); //Move right at speed1 once threshold met
delay(50);

} else if(val4>80) {

Mouse.move(xAxis5, yAxis,wheel); //Move right at speed2 once threshold met
delay(75);

} else if(val4>105) {
```

```
Mouse.move(xAxis6, yAxis,wheel); //Move right at speed3 once threshold met
delay(10);

Mouse.end();
delay(500);
}
}
```

References

1. Texas Instruments. INA128 data sheet, product information and support | TI.com [Internet]. www.ti.com. [cited 2023 Dec 15]. Available from: https://www.ti.com/product/INA128?utm_source=google&utm_medium=cpc&utm_campaign=asc-null-null-GPN_EN-cpc-pf-google-eu&utm_content=INA128&ds_k=INA128+Datasheet&DCM=yes&gad_source=1&gclid=Cj0KCQiAj_CrBhD-ARIsAliMxT-rjC8iHgNp9AHkec8-V4IqlEZvYL237-Z0p2kP-W_eH_Or5yOD5S8aAsgfEALw_wcB&gclidsrc=aw.ds
2. Texas Instruments. TL082 data sheet, product information and support | TI.com [Internet]. www.ti.com. [cited 2023 Dec 15]. Available from: <https://www.ti.com/product/TL082>
3. Texas Instruments. TL084 Datasheet [Internet]. Ti.com. 2021. Available from: <https://www.ti.com/product/TL084>
4. megcircuitsprojects. EMG Sensing Circuit [Internet]. Instructables. Available from: <https://www.instructables.com/EMG-Sensing-Circuit/>
5. IEEE. Influence of inter-electrode distance on EMG | IEEE Conference Publication | IEEE Xplore [Internet]. ieeeexplore.ieee.org. [cited 2023 Dec 15]. Available from: <https://ieeexplore.ieee.org/document/1020377>
6. Young AJ, Hargrove LJ, Kuiken TA. The Effects of Electrode Size and Orientation on the Sensitivity of Myoelectric Pattern Recognition Systems to Electrode Shift. IEEE Transactions on Biomedical Engineering. 2011 Sep;58(9):2537–44.
7. OpenStax. Muscles That Move the Forearm [Internet]. Wikimedia Commons. 2016 [cited 2023 Dec 15]. Available from: https://commons.wikimedia.org/wiki/File:1120_Muscles_that_Move_the_Forearm.jpg
8. Anatomy TV. Anatomy.tv | 3D Human Anatomy | Primal Pictures [Internet]. www.anatomy.tv. Available from: <https://www.anatomy.tv/titles>
9. Mouse - Arduino Reference [Internet]. www.arduino.cc. [cited 2023 Dec 15]. Available from: <https://www.arduino.cc/reference/en/libraries/mouse/>